

Supply initial density guess

$\rho_{\text{init}}(\vec{r})$ to Kohn Sham equations

Kohn-Sham method

$$v_{\text{ext},s}(\vec{r}) = v_H(\vec{r}) + v_{\text{xc}}(\vec{r}) + v_{\text{ext}}(\vec{r})$$

$$\hat{H}_{\text{KS}} = -\frac{\hbar^2}{2m} \vec{\nabla}^2 + v_{\text{ext},s}(\vec{r})$$

$$\hat{H}_{\text{KS}} \varphi_i(\vec{r}) = E_i \varphi_i(\vec{r})$$

$$\rho(\vec{r}) = \sum_{i=1}^n f_i |\varphi_i(\vec{r}_i)|^2$$

Convergence criterion satisfied?

No

Yes

Use $\rho_{\text{final}}(\vec{r})$ to minimize total energy functional

$$E_{V_{\text{ext}}}[\rho] = T_{e,s}[\varphi_i\{\rho\}] + V_{ee,H}[\rho] + E_{\text{xc}}[\rho] + V_{\text{el}}[\rho]$$